Questions:

Paper Work:

1- Find the mean and standard deviation of A.

A = {9,10,11,7,13, 7,4,11,13,10}

1) Find the mean and standard deviation of A.

$$A = \{9, 10, 11, 7, 13, 7, 4, 11, 13, 10\}$$

Ans) Given,

$$A = \{9, 10, 11, 7, 13, 7, 4, 11, 13, 10\}$$

To calculate mean we know that,

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \cdots x_n).$$

Here,

$$\bar{A} = \frac{1}{n}(A_1 + A_2 + A_3 + A_4 + \cdots A_n)$$

$$n = 10$$

$$\bar{A} = \frac{9+10+11+7+13+7+4+11+13+10}{10}$$

$$= \frac{95}{10} = 9.5$$

$\therefore$ Mean, is $\underline{9.5}$

Now, To calculate standard deviation,

$$\sigma = \sqrt{\frac{\Sigma (x_i - \mu)^2}{n-1}}$$

Here, $n = 10$, $\mu = 9.5$

$$\sigma = \sqrt{\frac{(9-9.5)^2+(10-9.5)^2+(11-9.5)^2+(7-9.5)^2+(13-9.5)^2+(7-9.5)^2+(4-9.5)^2+(11-9.5)^2+(13-9.5)^2+(10-9.5)^2}{(10-1)}}$$

$$= \sqrt{\frac{(-0.5)^2+(0.5)^2+(1.5)^2+(-2.5)^2+(3.5)^2+(-2.5)^2+(-5.5)^2+(1.5)^2+(3.5)^2+(0.5)^2}{9}}$$

$$= \sqrt{\frac{(0.25)+(0.25)+(2.25)+(6.25)+(12.25)+(6.25)+(30.25)+(2.25)+(12.25)+(0.25)}{9}}$$

$$= \sqrt{\frac{72.5}{9}} = \sqrt{8.055}$$

$$= 2.838$$

$\therefore$ standard deviation $\sigma = 2.838$

Python Work:

3- Write a code in Python:

a) **Enter vector A of question 1 in Python. Copy the code. (Hint: A<- c(2,4,…))**

b) **Find the mean and standard deviation of A. Copy the code.**

c) **Do your results match question 1?**

```
import statistics

# Define data vector A

A = [9, 10, 11, 7, 13, 7, 4, 11, 13, 10]

# Calculate the mean

mean_A = statistics.mean(A)

# Print the mean

print("Mean of A is %s" % mean_A)

# Calculate the standard deviation

std_deviation_A = statistics.stdev(A)

# Print the standard deviation

print("Standard Deviation of A is %s" % std_deviation_A)
```

```
In [1]:  import statistics

         # Define data vector A
         A = [9, 10, 11, 7, 13, 7, 4, 11, 13, 10]

         # Calculate the mean
         mean_A = statistics.mean(A)

         # Print the mean
         print("Mean of A is %s" % mean_A)

         # Calculate the standard deviation
         std_deviation_A = statistics.stdev(A)

         # Print the standard deviation
         print("Standard Deviation of A is %s" % std_deviation_A)

         Mean of A is 9.5
         Standard Deviation of A is 2.8382310609877335
```

Yes the results of mean and standard deviation match question 1.